

# Problem – Solution Fit Template

## Project Design Phase

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| <b>Project Name</b>      | Automated Network Request Management in ServiceNow |
| <b>Phase</b>             | Phase 3: Project Design                            |
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| <b>Platform / Domain</b> | ServiceNow / IT Service Management (ITSM)          |
| <b>Academic Year</b>     | 2025 – 2026                                        |

## Problem – Solution Fit Matrix

### 1. Customer Segment(s)

- Network Requesters: IT Staff, Software Developers, Systems Administrators.
- Approvers: Line Managers, Cybersecurity / IT Security Operations Leads.
- Fulfillment Operators: Network Engineers, System Administrators.
- Oversight: IT Administrators responsible for compliance and reporting.

### 2. Jobs-to-be-Done / Problems

- Request firewall policy changes, corporate VPN credentials, or VLAN reconfigurations rapidly.
- Ensure all infrastructure modifications undergo valid risk assessment and security vetting before fulfillment.
- Track request lifecycle status transparently without chasing engineers via email or phone.
- Generate compliance-ready audit reports for infrastructure change history.

### 3. Triggers

- A development team deploys a new service requiring an external port to be opened.

- A vendor or contractor being onboarded needs a secure, isolated corporate VPN profile.
- An upcoming security compliance deadline demands rapid deprecation of stale access rules.
- A network incident requires emergency IP or firewall rule reconfiguration.

#### 4. Before / After Emotional Journey

- Before: Frustrated by bureaucratic delays, anxious about missed deployment timelines, blind to request progression, and overwhelmed by informal multi-channel communication.
- After: Confident in a single submission portal, informed by real-time notifications, reassured by SLA visibility, and satisfied by structured, traceable outcomes.

#### 5. Proposed Solution

- Automated Network Request Management System built on ServiceNow: a structured Service Catalog with 8 network request types, dynamic validated intake forms, multi-stage Flow Designer approval workflows, automated task assignment, SLA tracking with escalation, and comprehensive audit logging — all accessible through the ServiceNow Service Portal.

#### 6. Solution Differentiators

- Uniquely combines client-side regex validation (IP/VLAN/port format enforcement) with risk-based conditional routing within a single ServiceNow PaaS environment — eliminating the need for external ticketing tools, custom code deployments, or middleware integrations.

#### 7. Measurable Outcomes

- Reduction in request MTTR by eliminating manual routing and follow-up cycles.
- Elimination of incomplete request submissions through mandatory field enforcement.
- 100% audit trail coverage for all infrastructure change requests.
- Improved SLA adherence through automated monitoring and escalation.
- Standardized request data enabling meaningful reporting and trend analysis.